SaprotHub: from a foundation model to a community

Democratizing protein language model training, sharing and collaboration

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Source note: full bioRxiv v7 (13 October 2025), checked against final journal supplementary information. Journal main text is subscription-only in this session.

The question: can biologists build on each other's models?

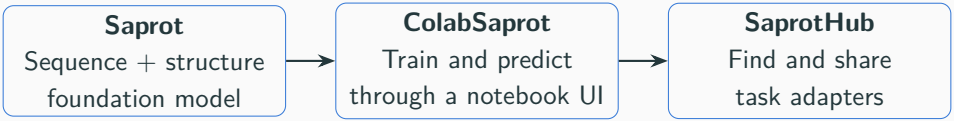
Protein language models can predict function, but adapting one to a new assay involves data processing, GPU training, evaluation and deployment.

The paper's proposed unit of collaboration
A shared foundation model plus small, task-specific adapters that researchers can train, reuse and combine.

1. Does the backbone support useful biological predictions?
2. Can lightweight training preserve predictive quality?
3. Can sharing improve predictions and reduce the expertise barrier?

Full preprint, introduction and Figs. 1–2. These are separate claims requiring separate evidence.

Three components have distinct responsibilities



OPMC is the broader community initiative. SaprotHub is its initial implementation, built on a shared Saprot backbone.

Running example
A lab measures enzyme activity for a small variant library, trains an adapter, and shares the adapter so another lab can predict or continue training.

Preprint, main text; Methods 1.12–1.14; final supplement, OPMC description. Example is illustrative.

A protein becomes a sequence of paired tokens

For residue i, combine amino acid s_i with Foldseek's discrete structural token f_i:

$$(s_1 f_1, \ldots, s_L f_L), \qquad s_i \in V, \quad f_i \in F.$$

Illustrative position	1	2	3	4
Amino acid	M	K	T	A
Structural token	d	c	p	y
Saprot token	Md	Kc	Tp	Ay
No reliable structure	M#	K#	T#	A#

There are 20 amino acids and 20 structural letters. Adding a mask symbol to each component gives $21 \times 21 = 441$ paired possibilities.

Preprint, Fig. 1a,b and Methods 1.1. Token sequence above is a teaching example, not a measured protein.

Pretraining learns residues in structural context

1. Start with approximately 40 million AlphaFold structures, filtered at 50% sequence identity.
2. Convert each structure to paired sequence–structure tokens.
3. Train a bidirectional Transformer to reconstruct selected masked tokens.

$$\texttt{Md\ \#c\ Tp\ Ay} \longrightarrow \text{predict the missing amino acid at site 2.}$$

Structure confidence matters: when pLDDT is below 70, replace the structural component with # in pretraining and downstream inputs.

What is being democratized?
Primarily adaptation and use. The 650M backbone itself required three months on 64 A100-80GB GPUs.

Preprint, Methods 1.2–1.4 and main text. Final supplement, Table 2.

Two prediction modes answer different questions

Zero-shot mutation scoring

No target-assay labels are required. Compare model probabilities of a mutant and wild-type residue given the masked context.

Useful for initial prioritization when no activity measurements exist.

Supervised adaptation

Inputs are sequences, optional structures, and measured labels. Train an adapter and prediction head for the desired assay.

Useful for learning a task-specific activity, localization or stability signal.

A model's evolutionary compatibility score is not a calibrated prediction of enzyme activity or clinical risk.

Preprint, Methods 1.6–1.8, 1.12. Last statement is journal-club interpretation.

A concrete zero-shot score

Let $p_i(a)$ be the predicted probability of amino acid a, summing over its possible structural tokens:

$$p_i(a) = \sum_{f \in F} P(x_i = af \mid x_{\setminus T}), \qquad S = \sum_{i \in T} \log \frac{p_i(s_i^{\text{mutant}})}{p_i(s_i^{\text{WT}})}.$$

T is the set of mutated sites; the model receives their masked context. The same wild-type structure is used for all variants.

Illustrative single mutation: K2R
If $p_2(\mathrm{R}) = 0.12$ and $p_2(\mathrm{K}) = 0.04$, then $S = \log 3 \approx 1.10$. The model favors R in that context; the assay must establish whether R increases activity.

Multi-site scores sum site contributions; they do not directly simulate a mutant's altered conformation.

Preprint, Methods 1.6–1.7, equations 1–2. Numerical example is invented for teaching.

LoRA makes task-specific models small enough to exchange

For a frozen weight matrix W, learn a low-rank update:

$$Wx \longrightarrow (W + BA)x, \qquad B \in \mathbb{R}^{d_{\text{out}} \times r}, \quad A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad r \ll d.$$

1. Keep the common Saprot backbone fixed.
2. Train lightweight adapter parameters and the task prediction head.
3. Exchange the adapter rather than a full copy of the large backbone.

The paper describes adapters as approximately 1% of the full model. Supplementary Fig. 4 finds broadly competitive performance with full fine-tuning across eleven tasks.

Preprint, Methods 1.12; final supplement, Fig. 4. Equation is a schematic of LoRA, not a claim about the exact rank/configuration.

The training workflow still requires scientific choices

1. Define the label: enzyme activity, fluorescence, localization, etc.
2. Prepare training, validation and test data; add structures if available.
3. Select a base or shared model, then fine-tune and monitor validation quality.
4. Predict on held-out data and inspect errors before sharing the adapter.

ColabSaprot provides installation, preprocessing, checkpoint selection, resumption and memory handling through notebook controls.

Our interpretation
The interface removes coding friction. It does not determine whether labels are comparable, the split is biologically meaningful, or deployment is outside the training distribution.

Preprint, Methods 1.9, 1.11 and 1.14; Supplementary Fig. 3.

Does structure help, beyond a different training dataset?

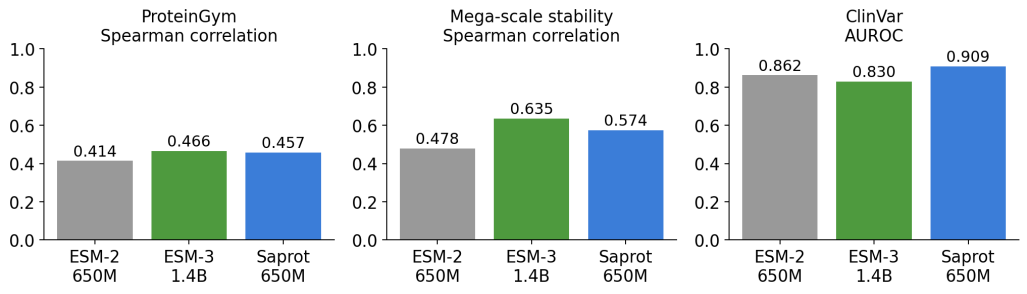
The controlled 35M comparison uses identical architecture, data and hyperparameters; Saprot-Seq masks every structural component.

30% structure-similarity split	ESM-2	Saprot-Seq	Saprot
Binding-site detection (MCC)	0.271	0.283	**0.434**
Metal binding (accuracy, %)	68.75	65.56	**72.22**
Structural similarity (ρ)	0.719	0.707	**0.754**
Thermostability (ρ)	0.618	0.623	**0.629**

These comparisons support a contribution from structural tokens. Gains differ substantially by task; they are not a universal multiplier.

Final journal supplement, Table 3. Primary 650M benchmarks mostly use a 70% LDDT split; sequence-only variant datasets retain original splits.

Zero-shot results: strong, but benchmark-dependent



Saprot improves over ESM-2 on all three reported benchmarks. ESM-3 is stronger on ProteinGym and Mega-scale, whereas Saprot is stronger on ClinVar in these comparisons.

Redrawn from final supplement, Table 7. Each metric has its own interpretation; AUROC is not a correlation. Dataset filtering is described in preprint Methods 1.7.

Supervised results support breadth, not uniform superiority

Task and metric	ESM-2	ESM-3	Saprot
Subcellular localization (%)	84.67	81.36	**86.18**
Binding-site detection (MCC)	0.540	0.548	**0.602**
Structure class (%)	79.90	**82.34**	80.80
Fluorescence (ρ), no structure	0.698	**0.700**	0.699
β-lactamase (ρ), no structure	0.904	**0.917**	0.890

With structural inputs, Saprot is a useful general-purpose backbone. Without structure, its performance is competitive but not consistently higher.

The list of "potentially supported" tasks is broader than the tasks actually benchmarked.

Final supplement, Tables 1, 6 and 8. Models: ESM-2/Saprot 650M; ESM-3 1.4B. Values are reported point estimates.

Inverse folding is a useful extension, with a tradeoff

Given a backbone, encode its 3Di sequence and hide every amino acid:

$$(\#f_1, \ldots, \#f_L) \longrightarrow \text{amino acid distributions at all sites.}$$

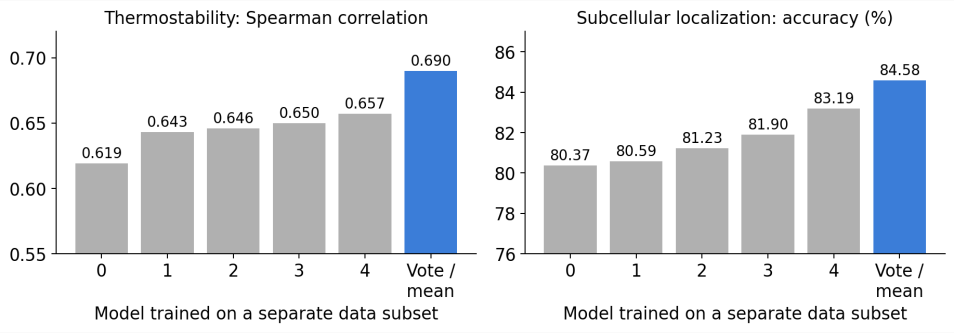
Saprot predicts sites simultaneously; ProteinMPNN uses autoregressive generation. The paper reports roughly 16-fold faster inference in its Fig. 1e setup.

100 recent structures	TM-score	RMSD (Å)	Recovery (%)
ProteinMPNN	0.88	1.77	48.47
Saprot, AF2 + CATH	0.87	2.02	45.19

Structural self-consistency is computational evidence; it does not establish folding or function experimentally.

Preprint, Fig. 1e and Methods 1.10; final supplement, Table 9 (December 2024 structures).

Sharing can help through an ordinary ensemble



Five models learn from separate random subsets of the same training set. Average regression outputs or vote over classifications. The combined prediction exceeds each component's mean score.

Redrawn from final supplement, Table 10; preprint Fig. 2e and Methods 1.13. Five training repetitions; test sizes 1,062 and 2,083. This simulates collaboration rather than measuring independent labs.

Continual training transfers task information

In Fig. 2f, the authors give a new user only 100, 200 or 500 training examples and compare:

1. An adapter initialized from the general pretrained backbone.
2. Continued adaptation from a task-specific shared model.

The shared initialization is stronger on thermostability and localization under these limited-data settings.

What this establishes
Reusing previous supervised learning can help when new labels are scarce. "From scratch" in this comparison means no previous task fine-tuning, not a randomly initialized protein model.

Open question: how much of the gain survives across laboratories, assay protocols or distant protein families?

Preprint, Fig. 2f and Methods 1.13. Cross-lab generalization is our discussion question.

The user study tests access under supported conditions

Twelve biology researchers without a coding/ML background receive datasets, instructions, GPU-enabled Colab accounts and three days. One ML PhD student provides the expert comparison.

Task, average score	Expert	Biologists
Thermostability (ρ)	0.664	0.658
Subcellular localization (%)	81.07	80.61
Stability (ρ)	0.592	0.607
eYFP fluorescence (ρ)	0.563	0.970

The eYFP comparison has unequal initialization: biologists reuse a task-trained shared model; the expert uses the general backbone. It demonstrates reuse, not superior training skill.

Preprint, Methods 1.15; final supplement, Table 11. Five participants per supervised task, 1,000 training samples each; volunteer self-selection acknowledged.

Wet-lab example 1: rank xylanase variants without labels

Workflow

Predict the Mth xylanase structure with AF3; generate structural tokens; score single substitutions with Saprot 650M; express selected variants in *Pichia pastoris*.

Measure reducing sugar release from xylan at 60°C, pH 5.0. Wild type is the reference.

Reported outcomes

13 of 20 tested variants improve activity.

R59S: 2.56× WT activity.

F212N: 1.89× activity; $>$ 60% retained after 4 h at 60°C, versus near-complete WT inactivation.

Limits: no random-selection comparator. Table 12 lists 22 candidates, with two signal-peptide variants excluded; its activity units are U/ml despite the "specific activity" label.

Preprint, Methods 1.16; final supplement, Table 12 and Figs. 6–7. Activity assays: at least three repeats; stability figure: two replicates.

Wet-lab example 2: improve a TDG-based editor

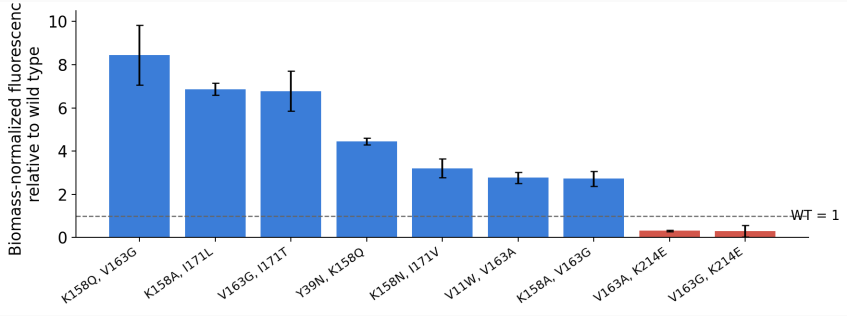
1. Score TDG single-site variants with sequence-only input using ColabSaprot v1 (650M).
2. Insert the 20 highest-ranked variants into an nCas9 fusion and transfect HeLa cells with Dicer-1 sgRNA.
3. Measure T-to-G substitutions by sequencing five days later.

Variant	WT	L74E	L74Q
T-to-G editing (%)	10.39	20.49	19.11
Fold change vs WT	1.00	1.97	1.84

The authors report 17/20 improvements; the displayed table has 16 values above WT. Treat that hit-count discrepancy as unresolved. One target site does not establish general editor improvement or specificity.

Preprint, Methods 1.18; final supplement, Table 15. Count discrepancy identified by inspection of reported values.

Wet-lab example 3: supervised selection of brighter GFP



A 35M model is fully fine-tuned on about 140,000 variants from four GFPs, then ranks 5 million candidate double mutants. Seven of the selected nine exceed WT fluorescence.

Redrawn from final supplement, Table 16 (reported uncertainties); preprint Methods 1.17. Biomass-normalized E. coli fluorescence, excitation/emission 488/520 nm. Two independent clones. This tests the backbone, not adapter-only training.

What should we conclude, and what needs a stronger test?

Supported by the evidence

Structure-aware tokens help several prediction tasks.

Small adapters can retain useful performance and make model reuse practical.

Non-ML users can complete tasks under the study's supported setup.

Still uncertain

Independent cross-lab adapter transfer and long-term community quality control.

Enrichment over random or expert variant selection in the wet-lab examples.

Formal privacy: sharing weights avoids direct data release, but does not prove protection against leakage.

Journal-club synthesis of preprint Figs. 1–2, Methods 1.13–1.18, and final supplement.

Discussion: what would make this useful in our lab?

1. Which assay has enough consistent labels to train a reusable adapter, and what held-out experiment would test transfer?
2. Would a task-matched shared adapter beat both a zero-shot backbone and a simple sequence-based predictor?
3. What metadata should travel with an adapter: assay units, split construction, protein coverage, backbone version and validation results?

Takeaway: the central contribution is an accessible path from foundation-model representations to reusable task-specific models. Its scientific value depends on validating the transfer, not only making the transfer easy.

Sources: Su et al., doi:10.1038/s41587-025-02859-7; full preprint doi:10.1101/2024.05.24.595648v7; final journal supplementary information. Source/access details in the accompanying README.